# Arista Access Node MAR-ACC-36 Alarm Resolution SOP - Failure Domain: Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss)

**Node:** MAR-ACC-36 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5996

## Description

The MAR-ACC-36 node, located in the Marathahalli Core, is an Arista access node that utilizes the EOS (Extensible Operating System) to manage network operations. The failure domain in question pertains to hardware/environmental faults encompassing ASIC memory parity errors, line card crashes, and power supply redundancy loss. These issues arise due to the inherent complexity of Arista's ASIC-based architecture, which integrates multiple components such as packet buffering, packet processing, and forwarding. The Access Node's role in this setup is critical, as it serves as the primary interface between the network and external devices. The likelihood of hardware faults is further amplified by the node's deployment in a high-traffic area such as the Marathahalli Core. Consequently, swift and accurate diagnosis, coupled with effective resolution, is essential to prevent network downtime and ensure continuous service availability.

## Pre-requisite Checks

- check: Verify the site location and confirm the node in question prior to initiating troubleshooting.

- check: Ensure that redundant link status is checked to verify link redundancy and confirm that the failure is not due to a link failure.

- check: Dispatch remote hands to Marathahalli Core, if necessary, to assist with troubleshooting and resolution efforts.

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity errors | High |
| Line card crashes | Medium |
| Power supply redundancy loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: ASIC memory usage, line card status, and power supply status, Description: This command will provide a snapshot of the node's system resources, including ASIC memory usage, line card status, and power supply status. This information will aid in identifying potential issues such as memory
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```
parity errors or line card crashes.

Command: show hardware linecard, Expected Output: Line card status, temperature, and voltage information, Description: This command will furnish detailed information about the line card's status, including temperature and voltage readings. This data will help determine if the issue is related to the line card itself or another component.

Command: show power redundancy, Expected Output: Power supply redundancy status, Description: This command will display the power supply redundancy status, indicating whether the node is operating in redundant or non-redundant mode. This information will aid in identifying potential issues related to power supply redundancy loss.
```

## Resolution Steps

Step 1: Step: Reset the line card, Command: reset hardware linecard, Description: If the line card is found to be malfunctioning, issue the 'reset hardware linecard' command to reset the line card. This will reboot the line card and potentially resolve the issue.

Step 2: Step: Replace the power supply, Command: replace power redundancy, Description: If the issue is related to power supply redundancy loss, replace the faulty power supply with a redundant one. This will ensure that the node continues to operate in redundant mode and prevent potential downtime.

Step 3: Step: Replace the ASIC, Command: replace hardware asic, Description: If the issue is related to ASIC memory parity errors, replace the faulty ASIC with a new one. This will ensure that the node continues to operate without memory parity errors and prevent potential downtime.

## Post-Check Verification

```
Command: show system resource, Expected Output: Normal system resource usage, Description: Verify that the system resource usage is normal after resolution.

Command: show hardware linecard, Expected Output: Normal line card status, Description: Verify that the line card status is normal after resolution.

Command: show power redundancy, Expected Output: Normal power supply redundancy status, Description: Verify that the power supply redundancy status is normal after resolution.
```

## Roll-back Procedures

- Step: Revert the line card change, Command: undo reset hardware linecard, Description: If the 'reset hardware linecard' command fails to resolve the issue, revert the change by issuing the 'undo reset hardware linecard' command. This will restore the original line card configuration.

- Step: Revert the power supply change, Command: undo replace power redundancy, Description: If the 'replace power redundancy' command fails to resolve the issue, revert the change by issuing the 'undo replace power redundancy' command. This will restore the original power supply configuration.

- Step: Revert the ASIC change, Command: undo replace hardware asic, Description: If the 'replace hardware asic' command fails to resolve the issue, revert the change by issuing the 'undo replace hardware asic' command. This will restore the original ASIC configuration.